

Cálculo Numérico - Aspectos Teóricos e Computacionais

Capítulo 1

Daniel Santos

May 22, 2026

Exercícios

1. Converta os seguintes números decimais para sua forma binária:

$$x = 37$$

$$y = 2345$$

$$z = 0.1217$$

2. Converta os seguintes números binários para sua forma decimal:

$$x = (101101)_2$$

$$y = (110101011)_2$$

$$z = (0.01101)_2$$

$$w = (0.111111101)_2$$

3. Seja um sistema de aritmética de ponto flutuante de quatro dígitos, base decimal e com acumulador de precisão dupla. Dados os números:

$$x = 0.7237 \times 10^4 \quad y = 0.2145 \times 10^{-3} \quad \text{e} \quad z = 0.2585 \times 10^1$$

efetue as seguintes operações e obtenha o erro relativo no resultado, supondo que x , y e z estão exatamente representados:

(a) $x + y + z$

(b) $x - y - z$

(c) x/y

(d) $(xy)/z$

(e) $x(y/z)$

Resolução

1. (a)

37

$$N_0 = 37 = 2 \cdot 18 + 1 \rightarrow a_0 = 1$$

$$N_1 = 18 = 2 \cdot 9 + 0 \rightarrow a_1 = 0$$

$$N_2 = 9 = 2 \cdot 4 + 1 \rightarrow a_2 = 1$$

$$N_3 = 4 = 2 \cdot 2 + 0 \rightarrow a_3 = 0$$

$$N_4 = 2 = 2 \cdot 1 + 0 \rightarrow a_4 = 0$$

$$N_5 = 1 = 2 \cdot 0 + 1 \rightarrow a_5 = 1$$

$$(37)_{10} = (100101)_2$$

(b)

2345

$$N_0 = 2345 = 2 \cdot 1172 + 1 \rightarrow a_0 = 1$$

$$N_1 = 1172 = 2 \cdot 586 + 0 \rightarrow a_1 = 0$$

$$N_2 = 586 = 2 \cdot 293 + 0 \rightarrow a_2 = 0$$

$$N_3 = 293 = 2 \cdot 146 + 1 \rightarrow a_3 = 1$$

$$N_4 = 146 = 2 \cdot 73 + 0 \rightarrow a_4 = 0$$

$$N_5 = 73 = 2 \cdot 36 + 1 \rightarrow a_5 = 1$$

$$N_6 = 36 = 2 \cdot 18 + 0 \rightarrow a_6 = 0$$

$$N_7 = 18 = 2 \cdot 9 + 0 \rightarrow a_7 = 0$$

$$N_8 = 9 = 2 \cdot 4 + 1 \rightarrow a_8 = 1$$

$$N_9 = 4 = 2 \cdot 2 + 0 \rightarrow a_9 = 0$$

$$N_{10} = 2 = 2 \cdot 1 + 0 \rightarrow a_{10} = 0$$

$$N_{11} = 1 = 2 \cdot 0 + 1 \rightarrow a_{11} = 1$$

$$(2345)_{10} = (100100101001)_2$$

(c)

0.1217

$k = 1$	$2r_1 = 0.2434 \Rightarrow d_1 = 0$	$r_2 = 0.2434$
$k = 2$	$2r_2 = 0.4868 \Rightarrow d_2 = 0$	$r_3 = 0.4868$
$k = 3$	$2r_3 = 0.9736 \Rightarrow d_3 = 0$	$r_4 = 0.9736$
$k = 4$	$2r_4 = 1.9472 \Rightarrow d_4 = 1$	$r_5 = 0.9472$
$k = 5$	$2r_5 = 1.8944 \Rightarrow d_5 = 1$	$r_6 = 0.8944$
$k = 6$	$2r_6 = 1.7888 \Rightarrow d_6 = 1$	$r_7 = 0.7888$
$k = 7$	$2r_7 = 1.5776 \Rightarrow d_7 = 1$	$r_8 = 0.5776$
$k = 8$	$2r_8 = 1.1552 \Rightarrow d_8 = 1$	$r_9 = 0.1552$
$k = 9$	$2r_9 = 0.3104 \Rightarrow d_9 = 0$	$r_{10} = 0.3104$
$k = 10$	$2r_{10} = 0.6208 \Rightarrow d_{10} = 0$	$r_{11} = 0.6208$
$k = 11$	$2r_{11} = 1.2416 \Rightarrow d_{11} = 1$	$r_{12} = 0.2416$
$k = 12$	$2r_{12} = 0.4832 \Rightarrow d_{12} = 0$	$r_{13} = 0.4832$
$k = 13$	$2r_{13} = 0.9664 \Rightarrow d_{13} = 0$	$r_{14} = 0.9664$
$k = 14$	$2r_{14} = 1.9328 \Rightarrow d_{14} = 1$	$r_{15} = 0.9328$
$k = 15$	$2r_{15} = 1.8656 \Rightarrow d_{15} = 1$	$r_{16} = 0.8656$
$k = 16$	$2r_{16} = 1.7312 \Rightarrow d_{16} = 1$	$r_{17} = 0.7312$
$k = 17$	$2r_{17} = 1.4624 \Rightarrow d_{17} = 1$	$r_{18} = 0.4624$
$k = 18$	$2r_{18} = 0.9248 \Rightarrow d_{18} = 0$	$r_{19} = 0.9248$
$k = 19$	$2r_{19} = 1.8496 \Rightarrow d_{19} = 1$	$r_{20} = 0.8496$
$k = 20$	$2r_{20} = 1.6992 \Rightarrow d_{20} = 1$	$r_{21} = 0.6992$
$k = 21$	$2r_{21} = 1.3984 \Rightarrow d_{21} = 1$	$r_{22} = 0.3984$
$k = 22$	$2r_{22} = 0.7968 \Rightarrow d_{22} = 0$	$r_{23} = 0.7968$
$k = 23$	$2r_{23} = 1.5936 \Rightarrow d_{23} = 1$	$r_{24} = 0.5936$
$k = 24$	$2r_{24} = 1.1872 \Rightarrow d_{24} = 1$	$r_{25} = 0.1872$

E assim segue infinitamente, não existindo representação exata.

2. (a)

$$\begin{aligned}
 & (101101)_2 \\
 &= 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\
 &= 32 + 0 + 8 + 4 + 0 + 1 = 45
 \end{aligned}$$

(b)

$$\begin{aligned}(110101011)_2 &= 1 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 \\ &\quad + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ &= 256 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 1 \\ &= 427\end{aligned}$$

(c)

$$\begin{aligned}(0.01101)_2 &= 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 0 \cdot 2^{-4} + 1 \cdot 2^{-5} \\ &= 0 + 0.25 + 0.125 + 0 + 0.03125 \\ &= 0.40625\end{aligned}$$

(d)

$$\begin{aligned}(0.11111101)_2 &= 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4} \\ &\quad + 1 \cdot 2^{-5} + 1 \cdot 2^{-6} + 1 \cdot 2^{-7} + 0 \cdot 2^{-8} + 1 \cdot 2^{-9} \\ &= 0.5 + 0.25 + 0.125 + 0.0625 + 0.03125 + 0.015625 \\ &\quad + 0.0078125 + 0 + 0.001953125 \\ &= 0.994140625\end{aligned}$$

3. (a) Valor de y desconsiderado por conta do acumulador de precisão dupla ter precisão apenas até a 8ª cada decimal.

$$\begin{aligned}\overline{x + y} &= 0.7237 \times 10^4 + 0.2145 \times 10^{-3} \\ &= 0.7237 \times 10^4 + 0.000000002145 \times 10^4 \\ &= 0.7237 \times 10^4 \\ \overline{(x + y) + z} &= 0.7237 \times 10^4 + 0.2585 \times 10^1 \\ &= 0.7237 \times 10^4 + 0.0002585 \times 10^4 \\ &= 0.7239585 \times 10^4\end{aligned}$$

$$\begin{aligned}\overline{x + y + z} &= 0.7239 \times 10^4, \text{ no truncamento.} \\ \overline{x + y + z} &= 0.7240 \times 10^4, \text{ no arredondamento.}\end{aligned}$$

O valor real seria $0.72395850214 \times 10^4$. Vamos calcular o erro relativo. Para simplificar, considerarei apenas o arredondamento.

$$\begin{aligned}
 EA_{x+y+z} &= |(x+y+z) - \overline{x+y+z}| \\
 &= |0.72395850214 \times 10^4 - 0.7240 \times 10^4| \\
 &= 0.00004149786 \times 10^4 = 0.4149786 \\
 ER_{x+y+z} &= \frac{EA_{x+y+z}}{\overline{x+y+z}} \\
 &= \frac{0.4149786}{0.7240 \times 10^4} \\
 &\approx 0.00005731748
 \end{aligned}$$

- (b) Valor de y desconsiderado por conta do acumulador de precisão dupla ter precisão apenas até a 8ª cada decimal.

$$\begin{aligned}
 \overline{x-y} &= 0.7237 \times 10^4 - 0.2145 \times 10^{-3} \\
 &= 0.7237 \times 10^4 - 0.000000002145 \times 10^4 \\
 &= 0.7237 \times 10^4 \\
 \overline{(x-y)-z} &= 0.7237 \times 10^4 - 0.2585 \times 10^1 \\
 &= 0.7237 \times 10^4 - 0.0002585 \times 10^4 \\
 &= 0.7234415 \times 10^4
 \end{aligned}$$

$\overline{x-y-z} = 0.7234 \times 10^4$, no arredondamento e truncamento.

O valor real seria $7237 - 0.0002145 - 2.585 = 7234.4147855$. Vamos calcular o erro relativo.

$$\begin{aligned}
 EA_{x-y-z} &= |(x-y-z) - \overline{x-y-z}| \\
 &= |7234.4147855 - 7234| \\
 &= 0.4147855 \\
 ER_{x-y-z} &= \frac{EA_{x-y-z}}{\overline{x-y-z}} \\
 &= \frac{0.4147855}{7234.4147855} \\
 &\approx 5.73 \times 10^{-5}
 \end{aligned}$$

(c)

$$\begin{aligned}\overline{x/y} &= (0.7237 \times 10^4)/(0.2145 \times 10^{-3}) \\ &= (0.7237/0.2145) \times 10^7 \\ &= 3.37389277389 \times 10^7 \\ &= 0.337389277389 \times 10^8\end{aligned}$$

$$\begin{aligned}\overline{x/y} &= 0.3373 \times 10^8, \text{ no truncamento.} \\ x/y &= 0.3374 \times 10^8, \text{ no arredondamento.}\end{aligned}$$

O valor real seria $7237/0.0002145 \approx 33738927.7389$.

$$\begin{aligned}EA_{x/y} &= |(x/y) - \overline{x/y}| \\ &= |33740000 - 33738927.7389| \\ &= 1072.2611 \\ ER_{x/y} &= \frac{EA_{x/y}}{\overline{x/y}} \\ &= \frac{1072.2611}{33738927.7389} \\ &\approx 3.17 \times 10^{-5}\end{aligned}$$

(d)

$$\begin{aligned}\overline{(xy)} &= 0.7237 \times 10^4 \cdot 0.2145 \times 10^{-3} \\ &= (0.7237 \cdot 0.2145) \times 10^1 \\ &= 0.15523365 \times 10^1 \\ &= 0.1552 \times 10^1 \\ \overline{(xy)/z} &= (0.1552 \times 10^1)/(0.2585 \times 10^1) \\ &= 0.60038684719 \times 10^0\end{aligned}$$

$$\begin{aligned}\overline{(xy)/z} &= 0.6003 \times 10^0, \text{ no truncamento.} \\ (xy)/z &= 0.6004 \times 10^0, \text{ no arredondamento.}\end{aligned}$$

O valor real seria ≈ 0.60051 .

$$\begin{aligned}
ER_{(xy)/z} &= \frac{EA_{(xy)/z}}{\overline{(xy)/z}} \\
&= \frac{0.60051 - 0.6004}{0.60051} \\
&\approx 1.83 \times 10^{-4}
\end{aligned}$$

(e)

$$\begin{aligned}
&\overline{(y/z)} \\
&= 0.2145 \times 10^{-3} / 0.2585 \times 10^1 \\
&= 0.82978723404 \times 10^{-2} \\
&= 0.8297 \times 10^{-2} \\
&\overline{(x(y/z))} \\
&0.8297 \times 10^{-2} \cdot 0.7237 \times 10^4 \\
&0.60045389 \times 10^2
\end{aligned}$$

$$\begin{aligned}
\overline{(x(yz))} &= 0.6004 \times 10^2, \text{ no truncamento.} \\
\overline{(x(yz))} &= 0.6005 \times 10^2, \text{ no arredondamento.}
\end{aligned}$$

$$\begin{aligned}
ER_{(x(y/z))} &= \frac{EA_{(x(y/z))}}{\overline{(x(y/z))}} \\
&= \frac{|0.60051 - 0.6005|}{0.60051} \\
&\approx 1.66 \times 10^{-5}
\end{aligned}$$